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Image Colorization with Regression and Classification

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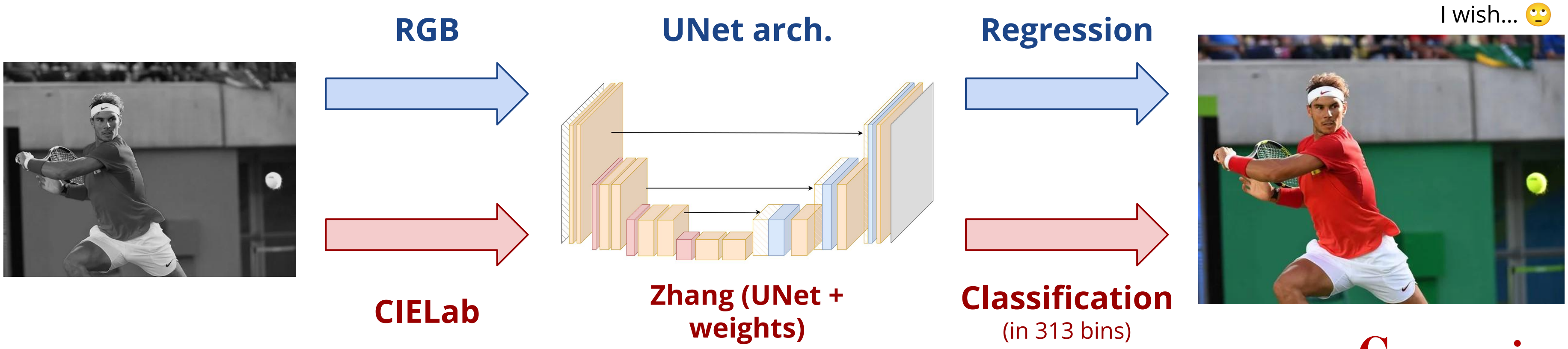
17644 - Applied Deep Learning

Image Colorization with Regression and Classification

The challenge is to **generate plausible and perceptually convincing colorizations** from **grayscale images**, following simplified versions of state-of-the-art architectures. These are the combinations used that require knowledge of **regression and classification** tasks and **color spaces in image processing:**



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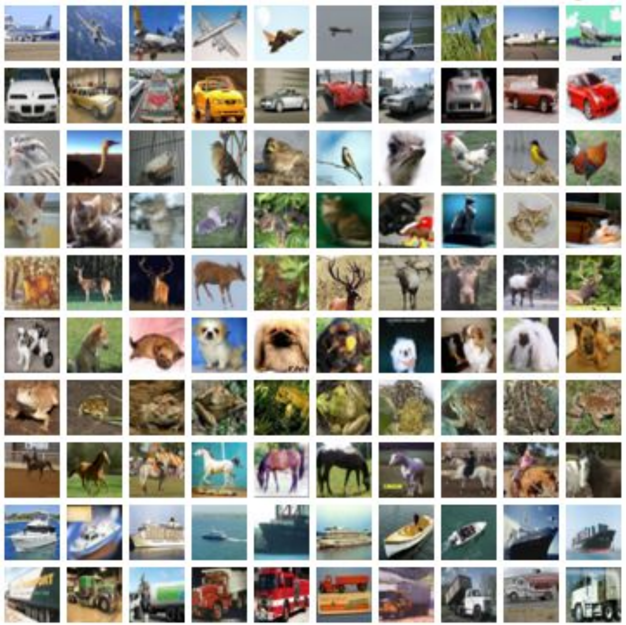
COCO Dataset and color spaces

For this project I have used a reduced version of the **COCO dataset**. Very used in research for several tasks:

- BBoxes
- Mask segmentation
- People keypoints
- Captioning tasks.

The version used has **30k images** and I have divided it **90%-10%** in a train - val split. Later, I have chosen several images to test visually the results.

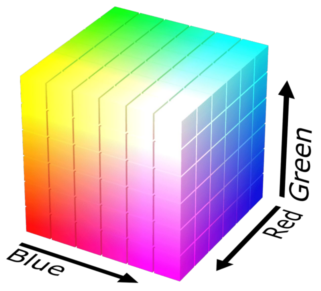


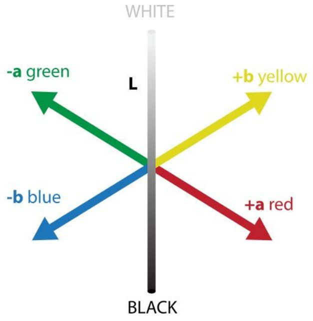
Feature Engineering

Since our task is image-to-image, for every entry, they are resized to (512, 512) or (256, 256) and create two images:

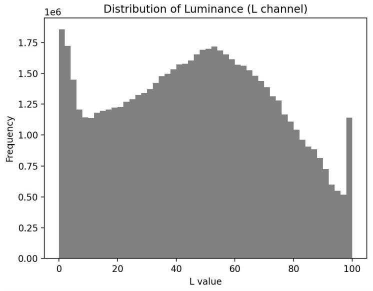
- **Input: greyscale** → (H, W, 1)
- **Target: color** → (H, W, 3*) same on both spaces

***Important:** Zhang's model takes the **grayscale L channel as input**, while the **ab channels** are **quantized into 313 bins** using **k-means clustering** with nearest neighbor encoding for classification.



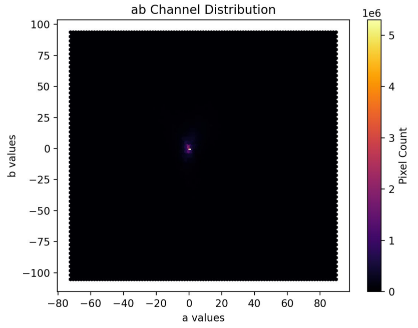


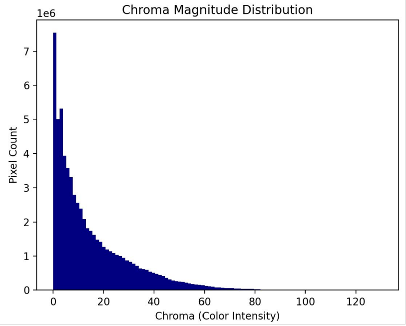
Exploratory Data Analysis

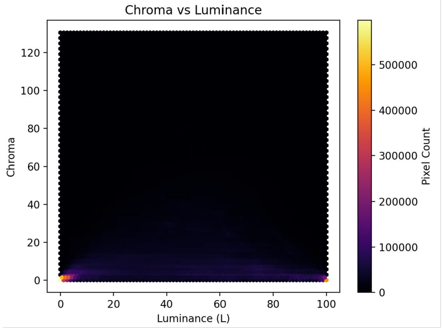


In general, human centered datasets are not very colorful, as it can be seen in the plots.

There is no clear relationship between luminescence and chroma, apart from having most of the pixels in the same bins.







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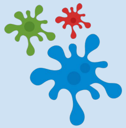
UNet Architecture - Modelling and Results

Basic Model

- ★ LeakyRelu, MAE
- ★ **(256, 256)**
- ★ **5 layers / 32 filters**
- ★ Dropout
- ★ Scheduler

Lets dig deeper...

- ★ **(512, 512)**
- ★ More scheduler
- ★ **1Conv: (k=7, s=2, p=3)**
- ★ **Clear U-shaped**



Loss, loss, loss

- ★ MAE + **Perceptual Loss**
- ★ Lambda = 0.3
- ★ VGG19 pretrained

Original Image

Predicted Image

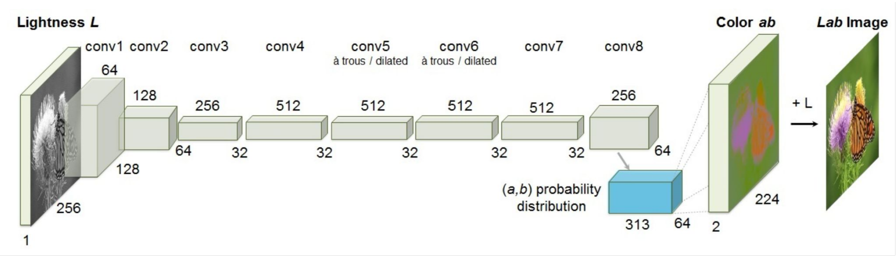
Original Image

Predicted Image

Original Image

Predicted Image

Zhang - Modelling and Results



Basic Model

- ★ sgd, momentum = 0.9
- ★ (256, 256)
- ★ 313 bins
- ★ Low batch size

Make weights weigh

- ★ Big sample for weights
- ★ Higher batch_size
- ★ adam

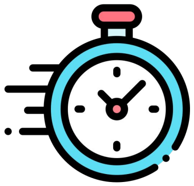
Original Image

Predicted Image

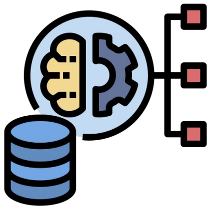
Original Image

Predicted Image

Main Insights

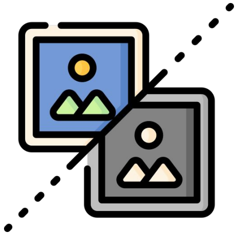


Regression is straightforward to implement and **faster to train (3x)**



Classification achieves **better results** and does **not overfit as easily.** Partly because of **weighted CrossEntropy** and balancing formula by Zhang.

Computer Vision tasks are **really heavy** to implement on personal laptops. Had to use **Lightning AI** (free tokens) to get access to GPUs. Also fine tune parameters like size, batch_size…



Upscaling tasks are very difficult to deploy. The model needs to **"learn" new information** from a **unique channel** and **output it to 3.**

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Thank you!